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Chapter 14
You Don't Have To Guess: No-Limit Bet Sizing

No-limit bet sizing is an important topic, but a topic that has generally been neglected from an analytical standpoint. We find recommendations in the existing literature ranging from pot-sized bets, to small underbets, to betting different amounts according to hand strength, and so on. These recommendations are usually based on a combination of experience and consensus. In some cases, however, we can find the analytical answer.

Let us first consider the half-street no-limit clairvoyant game. When we previously looked at the clairvoyant game, it was limit betting, and the clairvoyant player bet all his whirling hands and a fraction of the losers such that the other player was indifferent to calling. But in no-limit, there are a multitude of bet sizes to choose from. What makes one bet size better than another? How can we find the optimal strategies for such a game?

Example 14.1 - The Half-Street No-Limit Clairvoyance Game

One half street.
Pot size of 1 bet.
No-limit betting. (Stacks are n bets, where n is large)
Y is clairvoyant and his hand is randomly drawn from a distribution that is half winners and half dead hands.

Note: In limit games, we use the convention that the bet size is 1 and the pot is P units. However, in no-limit games with variable-sized bets, we instead use the convention that the pot is 1 unit, and the bet is *s*. Thus a bet of ***s*** units is a bet of s times the pot.

Let us first describe the structure of the strategy. After X checks dark, Y will clearly want to bet his winning hands for value, and protect those winning hands with bluffs such that X is indifferent to calling. Y will choose a bet size s that he will bet with both types of hands. If Y were to bet two different amounts, it would be necessary for such a strategy to be optimal that he have equal expectation against the nemesis with those two bet sizes. Then X will call in a proportion that makes Y indifferent to bluffing.

We can create a function *f(s),* which is Y's expectation from betting an amount. Then if we find the place where this function is maximum, we can find Y's proper bet size and have the solution to the game. When Y bets and bluffs appropriately, then X can fold or call at his discretion - both have equal expectation. Hence for finding Y's expectation, we can assume that X calls any bet as long as we hold Y's strategy constant. We will use this idea much more when we look at the [0,1] games later in Part III.

We know from the games we solved in Chapter 11 that Y's ratio of bluffs to bets will be α, and also that X will fold α of his hands that beat a bluff. Recall from Equation 11.2 that for no-limit games, we have an equivalent definition of α:

$\alpha = s/(1 + s)$

For any bet value *s*, Y will bluff $s/(1 + s)$ as many hands as he value bets. Suppose *y* is the fraction of Y's total hands that are winning hands.

$f(s) = p(\text{Y has nuts})\ (\text{one bet}) + p(\text{Y bluffs})\ (\text{lose one bet})$
$f(s) = (y)(s)+(ys/(1 + s))(-s)$
$f(s) = (y)(s)+-(ys^2)/(1+s)$

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$f(s) = ys/(1 + s)$

This function, as seen in the following graph, lacks any local maxima or minima:

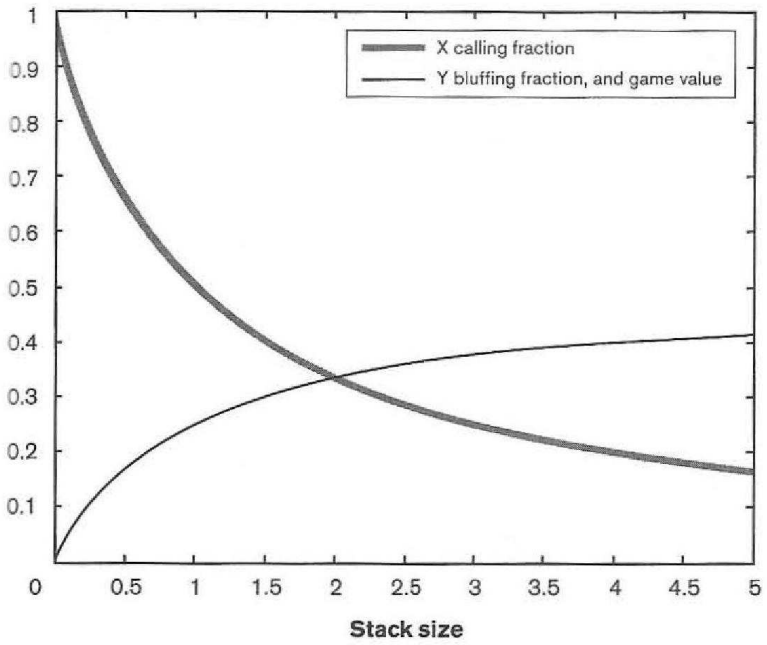

Figure 14.1. The half street clairvoyance game, y= 0.5

In fact, this function continues to increase as s goes to infinity. Hence Y maximizes his value by betting his whole stack and bluffing with the appropriate corresponding function.

We assumed in the above that Y's distribution was 50% nut hands and 50% dead hands. But suppose that y is $^3/_5$, meaning that $^3/_5$ Y's hands are winners and $^2/_5$ are bluff, and the stacks are just five pot-sized units. Now Y bets his whole stack. How should X respond? What if Y simply bets the pot or $s = 1$, how should X respond?

We consider the first question first. Well, the immediate answer based on the reasoning above would be that X should fold $\alpha = 5/(1 + 5) = {}^5/_6$ of his hands and call with the rest. This will make Y indifferent to bluffing. But to see why this is wrong, let's consider the strategy-pair we're discussing:

Y: Bet 5 with all winning hands and with all dead hands.

X: Call with $^1/_6$ of hands.

With a bluff, Y gains the pot of 1 for $^5/_6$ of the time for a total win of $^5/_6$, while losing the bet of $s = 5$ for $^1/_6$ a of the time, for a loss of $^5/_6$ of a unit, thus breaking even. With value bets, however, Y wins 5 units $^1/_6$ of the time, for a gain of $^5/_6$ unit per value bet. Since value bets make up $^3/_5$ of his hands overall, this is a total win of $^1/_2$ unit.

But what if X folded all the time? Then Y would win the entire pot with his bluffs, which make up $^2/_5$ of total hands, and gain nothing otherwise. So Y's equity would just be $^2/_5$ of a unit. So these cannot be the optimal strategies! X can unilaterally improve by simply folding all hands. Y could exploit this by bluffing all his hands. Suppose he does this. Then he would gain nothing with his value hands and the entire pot with this dead hands, for a total win of $^2/_5$ unit, Against

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this Y strategy, X's best response is to simply fold.

This is an important result; X's optimal strategy in this situation is to fold all his hands, even though Y is bluffing all of his dead hands. This is because Y's distribution is so strong that he can simply bet and "claim" the pot with all hands, because X loses money by calling at all. We will have more to say about these situations in the chapter on multi-street play. We have the game and equity graphs as follows:

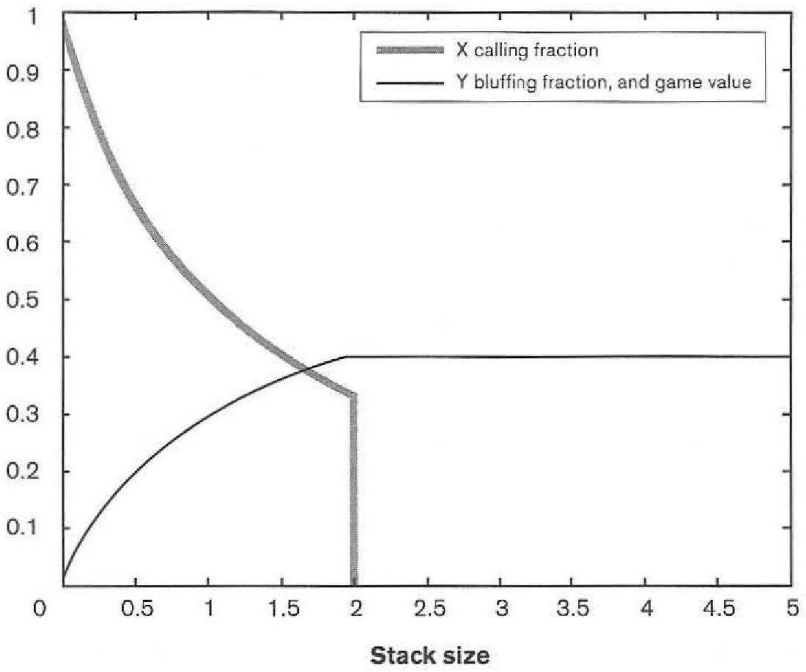

Figure 14.2. Half street no-limit clairvoyance game, y=0.6

Suppose Y doesn't bet all his chips, but simply bets the pot *(s =1)?* It might seem as though we can simply look at the above graph for s = 1 and see that we should call with half our hands. This will make him indifferent to bluffing a value s = 1, and so on. Unfortunately, it isn't that simple. The right answer is that X should ***still*** fold all hands. This is even true if Y had bet s= 1% of the pot.

To see why, consider how Y might exploit X's call. We know that X will fold all his hands if Y bets all his chips. Now suppose that X will respond to a smaller bet by calling in some mixed strategy. Y can exploit this by adopting a strategy of betting the smaller amount with his winning hands and betting all his chips with his bluffs. X could exploit that in turn, but since Y can increase his equity against X's strategy, it cannot be optimal. From another perspective, Y can simply push all-in with every hand and claim the pot-that is Y's optimal play. X does no better playing against Y's optimal strategy than he does by throwing his hand away in the dark. Hence X can play optimally by always throwing his hand away!

In Chapter 13, we considered the limit half-street AKQ game. There, we found that with a fixed pot in relation to the bet size, we could solve the game and find the proper betting and bluffing frequencies for both players. We continue our no-limit bet sizing investigations by considering the no-limit version of this game.

Example 14.2 - AKQ Game #3

One half street.
Pot size of 1 unit.

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No-limit betting. (Stacks are n bets, where n is large)

In this game, we have the same situation as previously. X checks dark, and Y then seeks to extract as much value from his hand distribution as possible. However, instead of a binary decision of "bet or check," Y is now faced with a more complex decision. If he does decide to bet, he must choose a bet amount as well. In fact, we can simply consider a "check" as betting zero units.

At a glance, we can see that it would be foolish for Y to bet more than the pot. If he does, then X picks off his bluffs with aces only and never pay off with kings. If he does this, he never loses money when Y has an ace, and gains more than the pot when Y bluffs a queen. We know from solving limit games previously that Y can profit by betting amounts smaller than the pot. Hence, we can discard ah bet amounts greater than the size of the pot.

Next, consider what would happen if we were to value bet a different amount than we bluffed. This would immediately be exploitable by X: X could simply fold kings to the value betting amounts and call the bluff amounts. For each value betting amount we utilize, we must have corresponding bluffs so that the ratio of bluffs to value bets (α) is preserved for that bet amount. One additional idea that is important here is that if we have a bet amount ***s*** such that we make more profit than betting some alternate amount t, then our strategies cannot be optimal. We could simply switch all the hands that bet t over to betting *s* and improve our equity. This means that either: 1) several bet amounts will have the same equity; or 2) there will be just one bet amount in Y's optimal strategy.

With all those ideas in mind, we can calculate the optimal strategies as follows. First we will find the optimal bluffing and calling frequencies for any bet amount *s*. These are the action frequencies which are X's best response to Y betting j and Y's proper bluffing frequency when betting *s*. Using these frequencies, we can solve for the expectation of the game, as a function of *s*. Once we do this, we can find the value of *j* for which Y's equity is maximized. Y will choose to bet and bluff at that particular value: X can do nothing to prevent this.

This is a common technique we use when considering games where different strategy choices can be made. Sometimes, we can solve the game or portions thereof in terms of a variable that only one player controls. When this is the case, we can find the strategies, calculate expectation, and then the player who controls that variable will choose its value such that this expectation is maximized.

Again we know from the games we solved in Chapter 11 that Y's ratio of bluffs to bets will be a, and also that X will fold α of his hands that beat a bluff, where $\alpha = s/(1 + s)$. (See Equation 11.2)

So for any bet value *s*, Y will bluff $s/(1 + s)$ of the time when he holds a queen. Likewise, X will fold a total fraction of his hands equal to $s/(1 + s)$.

Based on these frequencies, we can calculate the expectation of betting and bluffing any bet size *s*. We will create a ***value function*** *f(s)* that represents this expectation for any *s*.

We know from previous experience that Y will break even on his bluffs, because X will call often enough to make Y indifferent between bluffing and checking, and checking has an EV of zero. On his value bets, then, Y gains value when the first player calls with a king.

X must call with an overall frequency of 1 - α, or l/(1 + *s*). Subtracting $^1/_2$ from this expression for the aces with which X always calls:

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$c_{kings} = 2(1/(1 + s) - {}^1/_2)$
$c_{kings} = 2/(1 + s) - (1 + s)/(1 + s)$
$c_{kings} = (1 - s)/(1 + s)$

We find that the percentage of kings with which X will call is $(1 - s)/(1 + s)$. When he does call, Y's gain will be s units. So in the case where Y holds an ace and X holds a king, he gains $s(1 - s)/(1 + s)$ units. This case represents $^1/_6$ of the total hands, so the total expectation of the game to Y is:

$f(s) = s(1 - s)/6(1 + s)$

Y now chooses s such that the above expression is maximized. To do this requires a little calculus, but before we do that, you can see approximately where the peak will be by looking at the following graph:

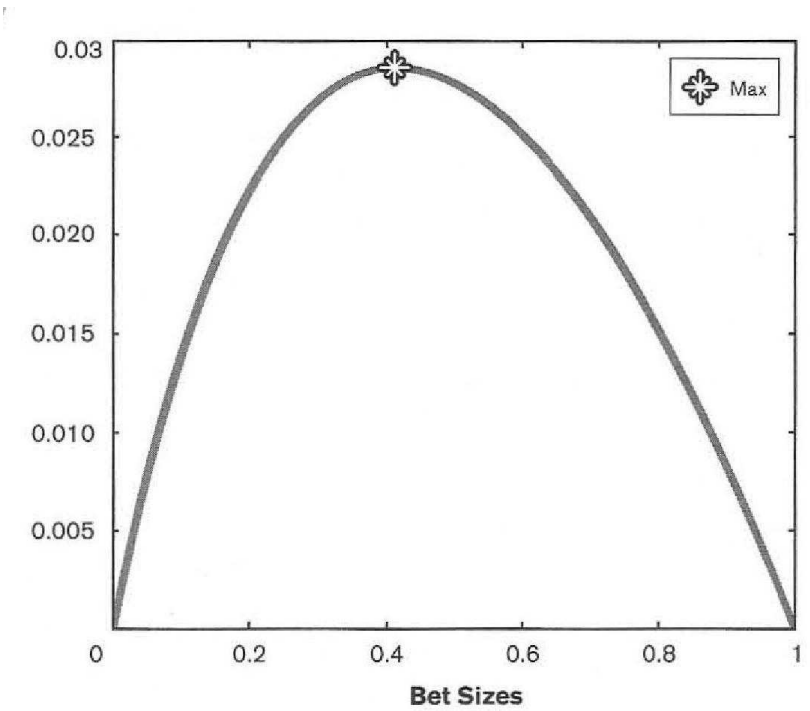

Figure 13.3. Equity of the NL clairvoyance game

Note: Remember that more complex derivations are marked off in boxes so that our less mathematical readers can just "take our word for it" if they choose.

We set the derivative of this expression equal to 0:

$f(s) = s(1 - s)/(1 + s)$
$f(s) = -(s^2 + 2s - 1)/(1+s)^2$
$s^2 + 2s - 1 = 0$
$(1+s)^2 = 2$
$s = -1 \pm \sqrt{2}$
$-1 - \sqrt{2}$ must be an extraneous solution, because we can't bet a negative amount.

So the value $\sqrt{2} - 1$ or approximately 0.114 is the proper amount to bet.

In his book *The Theory of Poker*, David Sklansky corned the phrase "Fundamental Theorem of Poker," a tongue-in-cheek reference to the Fundamental Theorem of Algebra and Fundamental

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Theorem of Calculus from introductory texts on those two subjects. The constant $\sqrt{2}$ - 1 appears so often in poker analysis that we will in the same vein go so far as to call it ***the golden mean of poker,*** and we call it *r* for short. We will see this value in a number of important results throughout this book.

$$r = \sqrt{2} - 1 \qquad \textbf{(14.1)}$$

The solution to this game is the first example we have seen of finding the proper bet amount in no-limit. This important concept revolves around the idea that we can find the expectation of the game for all different bet amounts. Once we have done this, we can find the bet amount value so that the bettor's expectation is maximized.

So in a game where the following conditions hold:

- You hold the nuts sometimes and a bluff sometimes.
- When you hold a bluff, your opponent holds the nuts half the time and the other half the time a hand that can only beat a bluff.

The optimal strategy is for you to bet $\sqrt{2}$ - 1 times the pot (a little more than 41%) with all of your nut hands, and to bluff with the corresponding frequency of bluffing hands

(1 - 1/$\sqrt{2}$). Your opponent in turn calls with all his nut hands and enough of his in-between hands such that his total calling frequency is 1/$\sqrt{2}$.

In addition to discrete games such as the AKQ game, we can use a similar idea to solve more complex no-limit games. Even continuous games such as the [0,1] game can be played no-limit. In the AKO game, we found that there was exactly one optimal bet size that Y used. In the half-street [0,1] game, however, we find that we occasionally bet all different bet sizes.

Example 14.3 - [0,1] Game #3

One half street.
Betting is no-limit; both players have arbitrarily large stacks.
The pot is 1 unit.
Folding is allowed.

Here, we have a similar game, where X checks dark, and Y tries to maximize his equity. However, we have a situation now where there are two variables that must be related - strength of hand and size of bet. In the limit AKQ game, if Y held a queen, he had a simple binary decision - bet or check. In the no-limit case, Y had to additionally choose what amount to bet.

In the half-street [0,1] no-limit game, Y must choose an amount to bet with each hand; it's possible that he may bet one particular amount, or vary his bet based on the strength of his hand. Just as in the previous game, we will attempt to solve the game in terms of the variables, and then have each player choose the variables he controls in order to maximize his equity.

We can first appeal to our previous investigations to conclude that Y will play a two-pronged strategy of betting and bluffing, and that his betting and bluffing regions will be related to each other by a ratio like α. So when X is faced with a bet of *t* from Y, he will need to call to make Y indifferent to bluffing at the edge of Y's bluffing threshold.

We can express X's strategy as a function of t. Let *x(t)* be X's calling threshold when faced with a

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bet of *t*. So *x(t)* will only consist of hands that can beat Y's bluffs - X will never call with hands that will lose even to Y's bluffs. The threshold where X will never call is x(0);. We will also call this value x_0.

X will choose his function *x(t)* such that if Y's handy is weaker than x_0, then Y is mdifferent to bluffing an amount *s*. In other words, X will call a bet of size *t* with exactly enough hands that Y is indifferent between betting *t* or checking for any value of t. So t_1, t_2, t_3 and so on, all have equity equal to checking. Hence Y is indifferent to betting any *s* value.

This leads to the following indifference for Y at *y*:

X's hand	**Y bet's**	**Y checks**
[0, x(s)]	-*s*	0
[x(s), y]	1	0

$y - x(s) - sx(s) = k$ (where *k* is constant)
$x(s) = (y - k)/(1+s)$

We know that $x(0) = x_0$, *so* $y - k = x_0$, *and therefore*

$$x(s) = x_0/(1+s) \qquad \textbf{(14.2)}$$

To recap, the above equation means that after X checks, if Y bets some amount *s*, X will call with hands better than $x_0/(1+s)$. At this point, we don't know what the value of x_0 is, only that X will never call with hands worse than that, no matter how little Y bets. We will solve for this value later.

Next we turn to Y's strategy.

In the same way, Y's value betting strategy will be a function of *s*, the bet amount that returns the hand value at which Y should bet a particular *s*.

If Y bets *s* for value at a particular hand value *y*, he will lose *s* when X holds a hand better than *y* this occurs with probability *y)* and he will win *s* when X holds a hand better than *x's* calling threshold *x(s)* but worse than *y*. This occurs with probability *(x(s) - y)*. Then his equity is:

<Y> = (chance that Y gets called and loses) *(-s)* + (chance that Y gets called and wins)(*s*)
$\langle Y\rangle = (y)(-s) + (x(s) - y)s$
$\langle Y\rangle = sx(s) - 2sy.$

Again using calculus, we can maximize this value:

$sx^2(s)+x(s)-2y=0$
$s(-x_0/(1 + s)^2) + x_0/(1+s)-2y=0$
$x_0/(1 + s)^2 - 2y = 0$

$y(s) = x_0/2(1 + s)^2$

This is Y's value betting strategy. Y's strategy bets different amounts at different Y values - in fact, bets a different amount at every Y value! The principle of information hiding would usually indicate that this is a terrible mistake because it can telegraph Y's hand. However, it turns out that in this particular game, X cannot make use of this information in an exploitive way, Y will

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match every value bet with a corresponding bluff in the proper ratio. Then, when faced with a bet of *s*, X can either call with confidence (when he knows that he has the best hand), call with uncertainty (knowing that Y either has a bluff which he can beat or a value bet which he cannot), or fold. Because X cannot raise, Y gives up no useful information by limiting the strength of his hand; X would always call with hands that beat Y.

Now the next part - matching up the bluffing region with the value betting region - is a little tricky; it again involves calculus. If you've understood what's happened up to this point, skipping the following section is fine. What we've done so far is show that the strategies for this game are as follows:

On [0, x(0)], X calls by his function *x(s)*. On [x(0),1], X folds.

On [0, y(0)], Y value bets by his function y(*s*). On [y(0),y'(0)]. Y checks behind. And with worse hands, Y bluffs according to y'(*s*).

In the following section, we will solve for the value of x_0 and show the relationship of the betting and bluffing region for Y.

We have a known value betting function:

$y(s) = x_0/2(1 + s)^2$

We can then construct a value betting differential; effectively the size of one infinitesimal movement along that function. Intuitively, this might be thought of as the size of the region in which we bet one particular amount. The *"dy"* is the movement of Y per every "*ds*' movement in s.

$dy = x_0/(1 + s)^2 ds$

It turns out that our familiar bluffing ratio of $s/(1 + s)$ applies to this differential.

So we have a bluffing differential:

$sx_0/ (1+s)^4 ds$

The total size of our bluffing region is then:

$$\int_0^\infty \frac{sx_0}{(1+s)^4} ds$$

Making the substitution $t = 1 + \boldsymbol{s}$, we have:

$$x_0 \int_{t=1}^{\infty} \frac{t-1}{t^4} dt$$
$$= x_0 \int_{t=1}^{\infty} \frac{1}{t^3} - \frac{1}{t^4} dt$$

$$= x_0[(-{}^1/_2\, t^2) + ({}^1/_3\, t^3)] \mid 1\infty$$

$$= {}^1/_6\, x_0$$

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The total size of the bluffing region is thus $^1/_6\, x_0$. Since the bluffing region begins immediately after x_0, we have:

$x_0 + {}^1/_6\, x_0 = 1$

So:

$x_0 = {}^6/_7$

Using the technique above, we can likewise map the bluffing region to the value betting region for Y; we leave this as an exercise for determined mathematicians. Instead, we return and state our strategy functions (Y's bluffing strategy as a function of s is omitted, but he bluffs some amount s with a total of $^1/_7$ of hands):

$x(s) = {}^6/_7(1+s)$
$y(s) = {}^3/_7(1+s)^2$

The important thing here, however, is not the particular strategies themselves, but the technique that we use for finding optimal bet sizes. In each of these two half-street games the AKQ and the [0,1]), we solve the game using variables to stand in for the unilateral strategy choices - in this case, the size of the bet that Y will make. Then the player who controls the variable selects its value such that his expectation is maximized.

We can learn many useful lessons from half-street games; over the past four chapters we have investigated a range of games:

- The clairvoyance game, which illustrated betting and bluffing ratios.
- The [0,1] Game #1, which introduced the concept of strategy regions and thresholds.
- The [0,1] Game #2, which showed the application of betting and bluffing ratios when the distributions are continuous and symmetrical.
- The [0,1] Jam-or-Fold games, which showed how uncertainty in the outcome of the
- showdown makes preflop jam-or-fold play much looser.
- The Jam-or-Fold no-limit game, which is a real-life solution to a game of some
- importance.
- The AKQ limit game, which again showed betting and bluffing ratios as well as introducing the idea of card removal.
- The AKQ no-limit game, which demonstrated the idea of calculating optimal bet size.
- The [0,1] game, which showed the much more complicated types of betting functions that arise from continuous distributions.

Key Concepts

- Bet sizing can be attacked analytically. To do this, we can first solve games for the strategies in terms of the bet size under consideration. Then the player who has control of the bet size maximizes his equity through his choice of a bet size.
- When one player has a very strong distribution or clairvoyance, situations can arise where the other player should simply give up no matter how small a bet the clairvoyant player makes.
- Information hiding is especially important when different bet sizes are available.

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